

Dexter Fernandes

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Education

Master's Degree in Robotics, *First-class degree*

Sep 2023 - Sep 2024

University of Bristol

- Engineered a UGV system (Clearpath Husky) with multi-sensor fusion (Velodyne LiDAR, Intel RealSense RGBD camera, Symetrica Verifinder).
- Implemented autonomous waypoint navigation using ROS2 Nav2 integrating LiDAR-based SLAM for precise, real-time environment mapping and navigation.
- Deployed real-time LiDAR SLAM and radionuclide detection on NVIDIA Jetson AGX Xavier edge hardware to generate dense 3D point cloud maps for precision radiological mapping running in a Docker container.
- Achieved precise radionuclide localisation using advanced odometry techniques like Scan-to-Map ICP, by significantly reducing drift in real-time LiDAR SLAM operations.
- Designed backend low-latency APIs (20ms) using FastAPI to enable real-time data streaming from edge devices (e.g., radiation sensors) and system telemetry.

Bachelor's Degree in Electrical and Electronics

Jul 2016 - May 2020

Visvesvaraya National Institute of Technology

- Secured sponsorships for Team Tesla (EV Racing) and designed/fabricated chassis for electric racing vehicles.

Work Experience

Esbaar, *AI Engineer*

Jan 2023 – Aug 2023

- Developed custom time-series models to classify 15 distinct oil rig operations while managing and analysing 3.6 million sensor data points.
- Optimised the hyperparameter tuning for time-series models by integrating MLOps, training 100+ models to boost F1-score by 39%.
- Led development of YOLOv5 vision models to enhance detection of rig workers and equipment, improving F1-score by 17%.
- Utilised Apache Kafka with Docker containers to transform a monolithic application into microservices, increasing fault-tolerance by 53%.
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.

LivNSense Technologies Pvt. Ltd, *Lead Software Engineer - AI*

May 2020 – Dec 2022

- Led a 4-engineer team to deliver ViCAS, a vision-based collision avoidance system, by curating domain-specific datasets, maintaining models in production, and resolving edge-case failures, reducing false positives by 27%.
- Deployed Intel OpenVINO-optimised vision models over Microsoft Azure IoT to rugged Intel VPU edge devices.
- Accelerated vision model inference with mixed-precision quantisation and problem-reframing, reducing latency by 72% (4.5 FPS → 15 FPS), enabling real-time reporting of OSHA violations.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.

Skills

- **Frameworks & Libraries:** ROS Nav2, PyTorch, TensorFlow, OpenCV, Sci-kit Learn, Intel OpenVINO
- **Computer Vision & Media Processing:** Point Cloud Optimization, Synthetic Training Data Generation, Object Detection, LiDAR SLAM
- **Backend & API Development:** FastAPI, REST APIs, Apache Kafka, PostgreSQL
- **DevOps & MLOps:** Git, Docker, CI/CD, Shell Scripting (Linux), MLFlow, Optuna, Microsoft Azure, DVC